Classical and Bayesian Estimation for Block Progressive Censored Competing Risks Data from the Kumaraswamy Distribution

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Abstract

This paper considers classical and Bayesian estimation for competing risks lifetime data collected under a block progressive censoring scheme. The cause-specific lifetimes follow Kumaraswamy distributions with a common first shape parameter and group- and cause-specific second shape parameters. Maximum likelihood estimates are developed without order restrictions and under two forms of order restriction, with existence and uniqueness established under stated conditions. Bayesian estimation is considered using an independent scale-invariant non-informative prior and a joint hierarchical Gamma model with proper informative Gamma hyperpriors. Posterior means are calculated when the corresponding first posterior moments are finite, and highest posterior density (HPD) intervals are constructed from proper posterior distributions. Approximate confidence intervals (ACIs) are considered for the maximum likelihood estimates. The performance of the estimators will be studied through simulation using absolute bias and mean-squared error, while interval estimates will be compared by average width and coverage probability. A bounded-support data set, or a lifetime data set transformed using a scientifically justified fixed upper bound, will be selected for illustration.

Keywords: Kumaraswamy distribution; block progressive censoring; competing risks; non-informative prior; maximum likelihood estimation; Bayesian estimation; order restrictions.

1 Introduction

Reliability testing is used to study the failure behaviour of products under specified operating conditions. Since testing every unit until failure can be expensive and time-consuming, censoring schemes are often used. Under progressive Type-II censoring, a fixed number of failures is observed and some surviving units are randomly removed after each observed failure.

In a block progressive censoring scheme, the test units are divided into groups and the groups are tested on separate facilities. Each group may have its own sample size, number of observed failures and removal plan. This setting allows differences between testing facilities to be studied. Ahmadi et al. (2018) considered block censoring schemes, while Wang et al. (2023) studied block progressive censored competing risks data.

In competing risks data, a unit may fail due to one of several causes, but only the first cause and its failure time are observed. We consider two independent causes of failure and assume that each cause-specific lifetime follows a Kumaraswamy distribution (Kumaraswamy, 1980; Jones, 2009). The paper develops classical estimation under unrestricted and order-restricted parameter

spaces. For Bayesian analysis, we use an independent scale-invariant non-informative prior and a joint hierarchical Gamma model. In the hierarchical model, the group-specific parameters have Gamma distributions conditional on cause-specific shape and rate parameters, and proper informative Gamma hyperpriors are assigned to those hyperparameters and to the common parameter. This specification permits posterior means and HPD intervals to be obtained by posterior sampling.

The paper is organised as follows. Section 2 describes the Kumaraswamy model, the block progressive censoring scheme and the likelihood. Section 3 presents unrestricted and order-restricted maximum likelihood estimation, together with approximate confidence intervals. Section 4 develops Bayesian estimation and HPD credible intervals. Section 5 outlines the simulation study and real-data analysis.

2 Model Description and Likelihood Function

2.1 The Kumaraswamy distribution

A random variable T follows a Kumaraswamy distribution with parameters $\alpha > 0$ and $\beta > 0$, written as $T \sim \text{Kum}(\alpha, \beta)$, if its density and distribution functions are

$$f(t) = \alpha\beta t^{\alpha-1}(1-t^{\alpha})^{\beta-1}, \qquad 0 < t < 1, \tag{1}$$

$$F(t) = 1 - (1-t^{\alpha})^{\beta}, \tag{2}$$

$$S(t) = (1-t^{\alpha})^{\beta}. \tag{3}$$

2.2 Competing risks model and BPCS data

Suppose that n units are divided into k groups. Group i contains n_i units and is tested on facility i. Under progressive Type-II censoring, m_i failure times are observed in group i, and

$$R_i = (r_{i1}, r_{i2}, \ldots, r_{im_i})$$

denotes its removal plan, where r_{ij} units are withdrawn after the jth observed failure and

$$\sum_{j=1}^{m_i}(r_{ij}+1) = n_i.$$

Let T_{i1} and T_{i2} be the lifetimes due to causes 1 and 2 in group i. We assume that these lifetimes are independent and

$$T_{is} \sim \text{Kum}(\alpha, \beta_{is}), \qquad i = 1, \ldots, k, \quad s = 1, 2,$$

where α is common to all groups and causes, while β_{is} varies with the group and cause. The observed lifetime and cause are

$$T_i = \min(T_{i1}, T_{i2}), \qquad \delta_i \in \{1, 2\}.$$

The observed data from group i are

$$\mathcal{D}_i = \{(x_{ij}, \delta_{ij}, r_{ij}) : j = 1, \ldots, m_i\},$$

where x_{ij} is the jth ordered failure time and δ_{ij} is its cause. Define

$$M = \sum_{i=1}^{k} m_i, \qquad e_{is} = \sum_{j=1}^{m_i} \mathbf{1}(\delta_{ij} = s), \qquad \tau_i = \beta_{i1} + \beta_{i2}.$$

By independence, the survival function of the observed lifetime in group i is

$$S_i(t) = (1 - t^{\alpha})^{\tau_i}.$$

The contribution of an observed failure due to cause s at time x_{ij} is

$$\alpha \beta_{is} x_{ij}^{\alpha-1} (1 - x_{ij}^{\alpha})^{\tau_i - 1}.$$

Each of the r_{ij} withdrawn units contributes a survival term. Hence, after omitting the censoring constant, the likelihood is

$$L(\boldsymbol{\beta}, \alpha; \mathcal{D}) \propto \alpha^{M} \prod_{i=1}^{k} \prod_{s=1}^{2} \beta_{is}^{e_{is}} \prod_{i=1}^{k} \prod_{j=1}^{m_i} \frac{x_{ij}^{\alpha-1}}{1 - x_{ij}^{\alpha}} \exp\left\{ - \sum_{i=1}^{k} \tau_i W_i(\alpha) \right\}, \tag{4}$$

where

$$W_i(\alpha) = - \sum_{j=1}^{m_i} (r_{ij} + 1) \ln(1 - x_{ij}^{\alpha}) > 0. \tag{5}$$

The log-likelihood is

$$\begin{aligned} \ell(\boldsymbol{\beta}, \alpha) = M \ln \alpha + \sum_{i=1}^{k} \sum_{s=1}^{2} e_{is} \ln \beta_{is} + (\alpha - 1) \sum_{i,j} \ln x_{ij} \\ - \sum_{i,j} \ln(1 - x_{ij}^{\alpha}) - \sum_{i=1}^{k} \tau_i W_i(\alpha). \end{aligned} \tag{6}$$

The group-level reliability functions used below are

$$S_i(t) = (1 - t^{\alpha})^{\tau_i}, \tag{7}$$

$$h_i(t) = \frac{\alpha \tau_i t^{\alpha-1}}{1 - t^{\alpha}}, \tag{8}$$

$$t_{i,0.5} = \left(1 - 2^{-1/\tau_i}\right)^{1/\alpha}. \tag{9}$$

Their maximum likelihood estimates are obtained by substitution, and their Bayesian estimates can be calculated from the corresponding posterior draws.

3 Maximum Likelihood Estimation

3.1 MLE under no order restriction

For a fixed value of α, differentiating the log-likelihood with respect to β_{is} and equating to zero gives

$$\widehat{\beta}_{is}(\alpha) = \frac{e_{is}}{W_i(\alpha)}, \qquad i = 1, \ldots, k, \quad s = 1, 2. \tag{10}$$

This is an interior estimate when $e_{is} > 0$. If a cause count is zero, the likelihood supremum is approached as the corresponding parameter tends to zero.

Substituting (10) into the log-likelihood gives the profile log-likelihood of α, up to an additive constant:

$$\begin{aligned} \ell_p(\alpha) = M \ln \alpha + (\alpha - 1) \sum_{i,j} \ln x_{ij} - \sum_{i,j} \ln(1 - x_{ij}^{\alpha}) \\ - \sum_{i=1}^{k} m_i \ln W_i(\alpha). \end{aligned} \tag{11}$$

The estimate $\widehat{\alpha}$ maximises this one-dimensional function. The remaining MLEs are then

$$\widehat{\beta}_{is} = \frac{e_{is}}{W_i(\widehat{\alpha})}.$$

For numerical calculation, put

$$\phi_{ij}(\alpha) = \frac{x_{ij}^{\alpha} \ln x_{ij}}{1 - x_{ij}^{\alpha}}.$$

Then $W_i'(\alpha) = \sum_j (r_{ij} + 1)\phi_{ij}(\alpha)$, and the profile score equation is

$$G(\alpha) = \ell_p'(\alpha) = \frac{M}{\alpha} + \sum_{i,j} \ln x_{ij} + \sum_{i,j} \phi_{ij}(\alpha) - \sum_{i=1}^{k} m_i \frac{W_i'(\alpha)}{W_i(\alpha)} = 0. \tag{12}$$

3.2 Existence and uniqueness of the unrestricted MLE

Assume $e_{is} > 0$ for every group and cause, and that the failure times are not all equal within at least one group. Put $y_{ij} = -\ln x_{ij} > 0$, $q(z) = -\ln(1 - e^{-z})$, and $n_i = \sum_j (r_{ij} + 1)$. Then

$$W_i(\alpha) = \sum_j (r_{ij} + 1) q(\alpha y_{ij}).$$

As $\alpha \downarrow 0$, $q(\alpha y) = -\ln \alpha - \ln y + o(1)$, so

$$W_i(\alpha) = n_i \ln(1/\alpha) + O(1), \qquad \frac{W_i'(\alpha)}{W_i(\alpha)} = -\frac{1 + o(1)}{\alpha \ln(1/\alpha)}.$$

Also, $\phi_{ij}(\alpha) = -1/\alpha + O(1)$. The terms M/α and $\sum_{i,j} \phi_{ij}(\alpha)$ therefore cancel to bounded order, while the last term in $G(\alpha)$ tends to $+\infty$. Hence $G(\alpha) \to +\infty$ as $\alpha \downarrow 0$.

For the other endpoint, let $y_i^* = \min_j y_{ij}$ and

$$D = \sum_{i=1}^{k} \sum_{j=1}^{m_i} (y_{ij} - y_i^*) > 0.$$

As $\alpha \to \infty$, $W_i(\alpha)$ is asymptotic to a positive constant times $e^{-\alpha y_i^*}$, so $W_i'/W_i \to -y_i^*$. Since $\phi_{ij}(\alpha) \to 0$,

$$\lim_{\alpha \to \infty} G(\alpha) = -D < 0.$$

Continuity and the intermediate value theorem give at least one root. The corresponding profile log-likelihood also tends to $-\infty$ at both endpoints: its leading behaviour is $-M \ln \ln(1/\alpha) + O(1)$ as $\alpha \downarrow 0$ and $-D\alpha + M \ln \alpha + O(1)$ as $\alpha \to \infty$.

For uniqueness, define $q(z) = -\ln(1 - e^{-z})$. Direct differentiation gives

$$q'(z) = -\frac{1}{e^z - 1}, \qquad q''(z) = \frac{e^z}{(e^z - 1)^2},$$

and hence

$$q(z)q''(z) - [q'(z)]^2 = \frac{e^z q(z) - 1}{(e^z - 1)^2} > 0.$$

The last inequality follows from $-\ln(1 - u) > u$ for $0 < u < 1$. Thus q is log-convex. A positive weighted sum of log-convex functions is log-convex, so each W_i is log-convex and $(\ln W_i)'' \geq 0$. Also,

$$0 < \phi_{ij}'(\alpha) = \frac{y_{ij}^2 e^{\alpha y_{ij}}}{(e^{\alpha y_{ij}} - 1)^2} = \frac{1}{\alpha^2} \left[\frac{\alpha y_{ij}}{2 \sinh(\alpha y_{ij}/2)} \right]^2 < \frac{1}{\alpha^2}.$$

Since there are M observed failure times,

$$G'(\alpha) = -\frac{M}{\alpha^2} + \sum_{i,j} \phi_{ij}'(\alpha) - \sum_i m_i (\ln W_i)'' < 0.$$

Therefore G is strictly decreasing. The endpoint limits established above have opposite signs, so G has exactly one root. The profile log-likelihood increases before this root and decreases after it; consequently, the unrestricted MLE exists and is unique under the stated conditions.

3.3 Approximate confidence intervals

Let $\boldsymbol{\theta} = (\beta_{11}, \beta_{12}, \ldots, \beta_{k1}, \beta_{k2}, \alpha)^T$. The observed information matrix is

$$\widehat{I} = -\left. \frac{\partial^2 \ell(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}\, \partial \boldsymbol{\theta}^T} \right|_{\boldsymbol{\theta} = \widehat{\boldsymbol{\theta}}}, \qquad \widehat{V} = \widehat{I}^{-1}.$$

The inverse observed information estimates the asymptotic covariance matrix of the MLE. A $100(1 - \gamma)\%$ approximate confidence interval for a parameter θ_j is

$$\widehat{\theta}_j \pm z_{1-\gamma/2} \sqrt{\widehat{V}_{jj}}. \tag{13}$$

For positive parameters, a log-scale interval can be used to avoid negative lower limits. For a reliability quantity $g(\boldsymbol{\theta})$, the delta method gives

$$\widehat{\mathrm{Var}}\{g(\widehat{\boldsymbol{\theta}})\} = \nabla g(\widehat{\boldsymbol{\theta}})^T \widehat{V}\, \nabla g(\widehat{\boldsymbol{\theta}}),$$

from which an approximate interval is obtained. When an order restriction is active, the ordinary normal ACI may not be valid at the boundary; a parametric bootstrap can be used for restricted MLEs.

3.4 MLE under order restriction

We consider the following restrictions separately:

- **Across groups:** $\beta_{1s} \le \beta_{2s} \le \cdots \le \beta_{ks}$ for each cause s.
- **Within groups:** $\beta_{i1} \le \beta_{i2}$ for every group i.

The cause-specific hazard is $h_{is}(t) = \alpha\beta_{is}t^{\alpha-1}/(1-t^{\alpha})$, so a larger β_{is} corresponds to a larger hazard for cause s. The across-group restriction is appropriate when the facilities or operating conditions are scientifically ordered in severity; the within-group restriction expresses the prior assumption that cause 2 is at least as hazardous as cause 1. These restrictions should be used only when justified by the application.

For the across-group restriction, fix α and calculate the unrestricted estimates $z_{is} = e_{is}/W_i(\alpha)$. Apply the pool-adjacent-violators algorithm (PAVA) separately for each cause, using $W_i(\alpha)$ as the weight for group i. If a consecutive block A is pooled, its common estimate is

$$\widetilde{\beta}_{is}(\alpha) = \frac{\sum_{i\in A} e_{is}}{\sum_{i\in A} W_i(\alpha)}, \qquad i \in A. \tag{14}$$

The restricted profile likelihood is obtained by substituting these estimates into the likelihood and maximising over α.

For the within-group restriction, the unrestricted estimates are retained whenever $e_{i1} \le e_{i2}$. If $e_{i1} > e_{i2}$, the two estimates are pooled:

$$\widetilde{\beta}_{i1}(\alpha) = \widetilde{\beta}_{i2}(\alpha) = \frac{m_i}{2W_i(\alpha)}. \tag{15}$$

The group total is unchanged. Hence, the restricted profile likelihood for α is the same as in the unrestricted case.

3.5 Existence and uniqueness under order restriction

Assume $e_{is} > 0$ for all groups and causes, and that the failure times are not all equal within at least one group. For fixed α, the log-likelihood is strictly concave in the cause-specific parameters. The inequalities define a closed convex order cone in the nonnegative orthant. Since all cause counts are positive, the log-likelihood tends to $-\infty$ if any component approaches zero; the unique constrained maximiser is therefore strictly positive. Hence the restricted estimates of the β_{is} exist and are unique for each fixed α.

For the within-group restriction, the two cause parameters in each group have the same weight $W_i(\alpha)$. Pooling changes the maximised contribution of that group only by a constant that does not depend on α; its total fitted parameter remains $m_i/W_i(\alpha)$. Thus the restricted profile log-likelihood differs from the unrestricted profile log-likelihood only by an additive constant. Its maximiser in α is therefore the same and is unique.

For the across-group restriction, let A denote a PAVA block for cause s, and put $E_{sA} = \sum_{i \in A} e_{is}$ and $W_A(\alpha) = \sum_{i \in A} W_i(\alpha)$. On any interval where the pooling blocks are fixed, the restricted profile score is

$$G_R(\alpha) = \frac{M}{\alpha} + \sum_{i,j} \ln x_{ij} + \sum_{i,j} \phi_{ij}(\alpha) - \sum_{s,A} E_{sA} \frac{W_A'(\alpha)}{W_A(\alpha)}.$$

Each W_A is a positive sum of log-convex functions and is therefore log-convex. Since the PAVA blocks partition the groups separately for each cause, $\sum_{s,A} E_{sA} = M$. Hence, within every interval on which the block structure is fixed,

$$G_R'(\alpha) = -\frac{M}{\alpha^2} + \sum_{i,j} \phi_{ij}'(\alpha) - \sum_{s,A} E_{sA} (\ln W_A)'' < 0.$$

It remains to check what happens when the PAVA blocks change. A transition occurs when two adjacent fitted block values become equal. If blocks A and B merge at such a point, then

$$\frac{E_{sA}}{W_A} = \frac{E_{sB}}{W_B} = c.$$

The score contribution before merging is $E_{sA} W_A'/W_A + E_{sB} W_B'/W_B = c(W_A' + W_B')$; after merging it is $(E_{sA} + E_{sB})(W_A' + W_B')/(W_A + W_B) = c(W_A' + W_B')$ as well. Thus the score is continuous at the transition. The same argument applies when a block splits. The transition equations are analytic in α. Unless two block values coincide identically, their crossing points are isolated; consequently, there are only finitely many non-identical transitions on any compact subinterval of $(0, \infty)$. If two fitted block values coincide identically, they may be treated as a single block without changing the fitted values or score. Therefore the piecewise strictly decreasing score is globally strictly decreasing. The restricted fitted parameters and restricted profile log-likelihood are continuous at these transitions.

Finally, the restricted profile log-likelihood is bounded above by the unrestricted profile log-likelihood, which tends to $-\infty$ at both endpoints. It therefore attains a maximum. Since its score is strictly decreasing wherever evaluated and continuous across transitions, the restricted profile is strictly concave and has only one maximiser. The across-group and within-group restrictions are treated separately; their simultaneous imposition is not covered here.

4 Bayesian Estimation

Under squared-error loss, the Bayes estimate of a parameter is its posterior mean. We consider two prior specifications: an independent scale-invariant non-informative prior and a joint hierarchical Gamma model with proper informative Gamma hyperpriors. The independent-prior estimates are available when the posterior and required moments are finite. The proper hierarchical prior provides a second Bayesian analysis and allows estimation of the common parameter, the

group-specific parameters and the hyperparameters. Order restrictions are considered separately for each prior specification.

4.1 Independent non-informative priors

For the positive parameters, we use the independent scale-invariant prior

$$\pi_I(\alpha, \boldsymbol{\beta}) \propto \frac{1}{\alpha} \prod_{i=1}^{k} \prod_{s=1}^{2} \frac{1}{\beta_{is}}, \qquad \alpha > 0, \quad \beta_{is} > 0. \tag{16}$$

Combining this prior with the likelihood gives

$$\pi_I(\boldsymbol{\beta}, \alpha \mid \mathcal{D}) \propto \alpha^{M-1} \prod_{i,j} \frac{x_{ij}^{\alpha-1}}{1 - x_{ij}^{\alpha}} \prod_{i=1}^{k} \prod_{s=1}^{2} \beta_{is}^{e_{is}-1} \exp\left\{ -\sum_{i=1}^{k} \tau_i W_i(\alpha) \right\}. \tag{17}$$

For fixed α, the conditional posterior distributions are

$$\beta_{is} \mid \alpha, \mathcal{D} \sim \text{Gamma}\big(e_{is}, W_i(\alpha)\big), \tag{18}$$

where the second argument denotes the rate. Integrating out the β_{is} gives the marginal posterior kernel

$$h_I(\alpha) = \alpha^{M-1} \prod_{i,j} \frac{x_{ij}^{\alpha-1}}{1 - x_{ij}^{\alpha}} \prod_{i=1}^{k} W_i(\alpha)^{-m_i}, \qquad \alpha > 0. \tag{19}$$

The posterior mean estimates are therefore

$$\begin{aligned} \widehat{\alpha}_{B,I} &= \frac{\displaystyle\int_0^\infty \alpha h_I(\alpha)\, d\alpha}{\displaystyle\int_0^\infty h_I(\alpha)\, d\alpha}, \\ \widehat{\beta}_{is,B,I} &= \frac{\displaystyle\int_0^\infty \frac{e_{is}}{W_i(\alpha)} h_I(\alpha)\, d\alpha}{\displaystyle\int_0^\infty h_I(\alpha)\, d\alpha}. \end{aligned} \tag{20}$$

Thus, the conditional mean $e_{is}/W_i(\alpha)$ is averaged over the posterior uncertainty in α to obtain the Bayes estimate of β_{is}. These integrals can be evaluated by one-dimensional numerical quadrature. In the unrestricted model, posterior draws can also be generated by first drawing α from the normalized marginal density proportional to $h_I(\alpha)$ and then drawing the β_{is} independently from their Gamma conditional distributions in (18). This marginal-then-conditional procedure applies to the unrestricted model; the order-restricted sampler is given separately below. The prior in (16) is improper, so posterior propriety must be checked. Put $y_{ij} = -\ln x_{ij}$, $y_i^* = \min_j y_{ij}$, and

$$D = \sum_{i=1}^{k} \sum_{j=1}^{m_i} (y_{ij} - y_i^*).$$

When every $e_{is} > 0$, integration over the β_{is} is finite. To check the remaining integral, note that as $\alpha \downarrow 0$, $1 - x_{ij}^{\alpha} \sim \alpha y_{ij}$ and $W_i(\alpha) = n_i \ln(1/\alpha) + O(1)$. Thus

$$h_I(\alpha) \asymp \frac{1}{\alpha\{\ln(1/\alpha)\}^M}.$$

This is integrable at zero exactly when $M > 1$. As $\alpha \to \infty$, $W_i(\alpha)$ is asymptotic to a positive constant times $e^{-\alpha y_i^*}$, giving

$$h_I(\alpha) \asymp \alpha^{M-1} e^{-D\alpha}.$$

This is integrable at infinity when $D > 0$, which means that at least one group has non-identical observed failure times. Hence the independent-prior posterior is proper if $e_{is} > 0$ for all i, s, $M > 1$, and $D > 0$. Under these conditions the posterior mean of α is finite. For the squared-error Bayes estimate of β_{is}, the additional tail integral contains $h_I(\alpha)/W_i(\alpha)$ and is finite under the sufficient condition

$$D > y_i^*, \qquad i = 1, \ldots, k.$$

These conditions should be checked for the dataset before reporting the corresponding posterior means. They are also sufficient after conditioning on an order restriction having positive posterior probability, since restriction to an event of positive probability preserves any finite first moment.

4.2 Joint hierarchical Gamma prior with informative hyperpriors

For the second Bayesian analysis, we use a joint hierarchical Gamma model. Conditional on cause-specific shape and rate parameters, let

$$\beta_{is} \mid a_s, b_s \sim \text{Gamma}(a_s, b_s), \qquad i = 1, \ldots, k, \quad s = 1, 2, \tag{21}$$

independently, where the second argument denotes the rate. To obtain a proper joint prior, assign proper informative Gamma hyperpriors:

$$\begin{aligned} &\alpha \sim \text{Gamma}(A_\alpha, B_\alpha), \\ &a_s \sim \text{Gamma}(A_s, B_s), \qquad b_s \sim \text{Gamma}(C_s, D_s), \qquad s = 1, 2, \end{aligned} \tag{22}$$

independently, with all hyperparameters positive and fixed. These constants are selected using prior knowledge and should be examined through a sensitivity analysis. Under the shape–rate parameterisation, for example, the prior means are A_α/B_α, A_s/B_s and C_s/D_s, respectively. Conditional independence of the β_{is} given (a_s, b_s), together with the shared hyperparameters, induces dependence between group parameters after the hyperparameters are integrated out.

Combining the likelihood with these priors gives the joint posterior kernel

$$\begin{aligned} &\pi_H(\alpha, \boldsymbol{\beta}, \boldsymbol{a}, \boldsymbol{b} \mid \mathcal{D}) \\ &\quad \propto \alpha^{M + A_\alpha - 1} e^{-B_\alpha \alpha} \prod_{i,j} \frac{x_{ij}^{\alpha - 1}}{1 - x_{ij}^{\alpha}} \\ &\qquad \times \prod_{s=1}^{2} \left\{ \frac{a_s^{A_s - 1} e^{-B_s a_s}}{\Gamma(a_s)^k} b_s^{k a_s + C_s - 1} e^{-D_s b_s} \prod_{i=1}^{k} \beta_{is}^{a_s + e_{is} - 1} \exp\left[-\sum_{i=1}^{k} (b_s + W_i(\alpha)) \beta_{is} \right] \right\}. \end{aligned} \tag{23}$$

The constants in the normalised Gamma densities that do not depend on unknown parameters have been omitted. In particular, the factor $\Gamma(a_s)^{-k}$ is retained because a_s is unknown.

The full conditional distributions for the group parameters and rate hyperparameters are

$$\beta_{is} \mid \text{rest}, \mathcal{D} \sim \text{Gamma}\big(a_s + e_{is},\ b_s + W_i(\alpha)\big), \tag{24}$$

$$b_s \mid \text{rest}, \mathcal{D} \sim \text{Gamma}\left(ka_s + C_s,\ D_s + \sum_{i=1}^{k} \beta_{is}\right). \tag{25}$$

The conditional density of the shape hyperparameter is not a standard distribution. Its kernel is

$$\pi(a_s \mid \text{rest}, \mathcal{D}) \propto a_s^{A_s - 1} e^{-B_s a_s} \frac{\left(b_s^k \prod_{i=1}^{k} \beta_{is}\right)^{a_s}}{\Gamma(a_s)^k}, \qquad a_s > 0, \tag{26}$$

and can be sampled using a one-dimensional slice sampler. The conditional density of the common parameter is

$$\pi(\alpha \mid \text{rest}, \mathcal{D}) \propto \alpha^{M + A_\alpha - 1} e^{-B_\alpha \alpha} \prod_{i,j} \frac{x_{ij}^{\alpha - 1}}{1 - x_{ij}^{\alpha}} \exp\left\{-\sum_{i=1}^{k} \tau_i W_i(\alpha)\right\}, \tag{27}$$

which can also be sampled by one-dimensional slice sampling on $(0, \infty)$. The earlier flat hyperpriors $\pi(a_s, b_s) \propto 1$ led to an improper joint posterior; they are replaced here by proper informative Gamma hyperpriors. To check propriety, integrate out β_{is}. For each cause, the resulting factor is

$$\prod_{i=1}^{k} \frac{\Gamma(a_s + e_{is})}{\Gamma(a_s)} \left(\frac{b_s}{b_s + W_i(\alpha)}\right)^{a_s} (b_s + W_i(\alpha))^{-e_{is}}.$$

Since e_{is} is a non-negative integer, $\Gamma(a_s + e_{is})/\Gamma(a_s)$ is a polynomial in a_s, while the ratio in parentheses is at most one and $(b_s + W_i)^{-e_{is}} \le W_i^{-e_{is}}$. The proper Gamma hyperpriors on a_s and b_s therefore give a finite bound proportional to $\prod_i W_i(\alpha)^{-m_i}$ after both causes are included. Near $\alpha = 0$, the remaining integrand is bounded by a constant times $\alpha^{A_\alpha - 1}\{\ln(1/alpha)\}^{-M}$, which is integrable because $A_\alpha > 0$. As $\alpha \to \infty$, it is bounded by a polynomial in α times $\exp[-(B_\alpha + D)\alpha]$, where $D \ge 0$ is defined in the independent-prior discussion. Hence the joint posterior is proper for observations in $(0, 1)$ and positive fixed Gamma prior shape and rate constants. For a finite posterior mean of β_{is}, a sufficient additional condition is $B_\alpha + D > y_i^*$; HPD intervals require posterior propriety but not finite posterior means.

Under squared-error loss, the Bayes estimates are the marginal posterior means

$$\widehat{\alpha}_{B,H} = E(\alpha \mid \mathcal{D}), \qquad \widehat{\beta}_{is,B,H} = E(\beta_{is} \mid \mathcal{D}). \tag{28}$$

The conditional mean of a group parameter is

$$E(\beta_{is} \mid \alpha, a_s, b_s, \mathcal{D}) = \frac{a_s + e_{is}}{b_s + W_i(\alpha)}.$$

Consequently, its marginal Bayes estimate may be evaluated by averaging this expression over the joint posterior of (α, a_s, b_s), or by averaging the sampled β_{is} values directly. With N' retained

posterior draws, the Monte Carlo estimates are

$$\widehat{\alpha}_{B,H} = \frac{1}{N'}\sum_{q=1}^{N'} \alpha^{(q)},$$
$$\widehat{\beta}_{is,B,H} = \frac{1}{N'}\sum_{q=1}^{N'} \beta_{is}^{(q)}, \qquad \widehat{a}_{s,B} = \frac{1}{N'}\sum_{q=1}^{N'} a_s^{(q)}, \quad \widehat{b}_{s,B} = \frac{1}{N'}\sum_{q=1}^{N'} b_s^{(q)}. \tag{29}$$

These expressions give separate posterior mean estimates for α and every β_{is}; numerical values are obtained after running the sampler on the selected data set.

For reproducibility, the unrestricted hierarchical sampler proceeds as follows: (i) initialise all parameters at positive values; (ii) update each β_{is} from (24); (iii) update each b_s from (25); (iv) update each a_s from (26) using slice sampling; and (v) update α from (27) using slice sampling. Repeat these steps for multiple chains, discard warm-up draws, and check trace plots, split-$\widehat{R}$, effective sample sizes and Monte Carlo standard errors before calculating posterior summaries. The prior hyperparameters should be varied in a sensitivity analysis to assess the effect of the informative prior choices.

4.3 Bayesian estimation under order restriction

Let $\mathcal{C}$ denote either the across-group or within-group restriction from Section 3, considered separately. The restricted posterior is obtained by conditioning the corresponding unrestricted posterior on $\boldsymbol{\beta} \in \mathcal{C}$:

$$\pi_{\mathcal{C}}(\cdot \mid \mathcal{D}) \propto \pi(\cdot \mid \mathcal{D})\, \mathbf{1}(\boldsymbol{\beta} \in \mathcal{C}). \tag{30}$$

Independent scale-invariant prior. For fixed α, each unrestricted full conditional is the Gamma distribution in (18). Under the across-group restriction, update β_{is} from this Gamma distribution truncated to

$$(L_{is}, U_{is}), \qquad L_{is} = \begin{cases} 0, & i = 1, \\ \beta_{i-1,s}, & i > 1, \end{cases} \quad U_{is} = \begin{cases} \beta_{i+1,s}, & i < k, \\ +\infty, & i = k. \end{cases}$$

Under the within-group restriction, use the truncated conditionals

$$\beta_{i1} \mid \text{rest}, \mathcal{D}, \mathcal{C} \sim \text{Gamma}(e_{i1}, W_i(\alpha)) \text{ truncated to } (0, \beta_{i2}),$$
$$\beta_{i2} \mid \text{rest}, \mathcal{D}, \mathcal{C} \sim \text{Gamma}(e_{i2}, W_i(\alpha)) \text{ truncated to } (\beta_{i1}, \infty).$$

The Gamma distributions use the shape–rate parameterisation. These coordinate-wise truncated conditionals are the full conditionals of the independent-prior posterior restricted to $\mathcal{C}$.

The conditional posterior of α, given the current values of $\boldsymbol{\beta}$, is

$$\boxed{\begin{aligned} \pi_I(\alpha \mid \boldsymbol{\beta}, \mathcal{D}, \mathcal{C}) \propto \alpha^{M-1} \prod_{i,j} \frac{x_{ij}^{\alpha-1}}{1 - x_{ij}^{\alpha}} \\ \times \exp\left\{ -\sum_{i=1}^{k} (\beta_{i1} + \beta_{i2}) W_i(\alpha) \right\}, \qquad \alpha > 0. \end{aligned}} \tag{31}$$

Because the current $\boldsymbol{\beta}$ already satisfies $\mathcal{C}$, the indicator in (30) is constant during this α update. The conditional density in (31) is therefore the same as the unrestricted conditional density given $\boldsymbol{\beta}$; it is not the unrestricted marginal density $h_I(\alpha)$.

In particular, the restricted marginal density has the form

$$h_{I,\mathcal{C}}(\alpha) \propto h_I(\alpha)\, P_I(\boldsymbol{\beta} \in \mathcal{C} \mid \alpha, \mathcal{D}). \tag{32}$$

For across-group ordering, the probability factor generally depends on α through the group-specific rates $W_i(\alpha)$. For the within-group ordering $\beta_{i1} \leq \beta_{i2}$, the two independent Gamma conditionals in a given group have the same rate $W_i(\alpha)$; scaling both variables by this common rate shows that the ordering probability is independent of α. Thus, for this particular within-group restriction and this independent prior, the marginal density of α remains proportional to $h_I(\alpha)$. The conditional sampler below is valid for either restriction.

The independent-prior restricted sampler is:

1. Initialise $\alpha > 0$ and all β_{is} at values satisfying the selected order restriction.
2. Update each β_{is} in turn from its truncated Gamma full conditional, using the current neighbouring parameter values as truncation limits.
3. Update α from (31), for example by one-dimensional slice sampling on $(0, \infty)$.
4. Repeat the updates, discard warm-up iterations, and retain the posterior draws after checking trace plots, split-$\widehat{R}$, effective sample sizes and Monte Carlo standard errors.

The restricted independent-prior Bayes estimates under squared-error loss are the posterior sample means

$$\widehat{\alpha}_{B,I,\mathcal{C}} = \frac{1}{N'} \sum_{q=1}^{N'} \alpha^{(q)}, \qquad \widehat{\beta}_{is,B,I,\mathcal{C}} = \frac{1}{N'} \sum_{q=1}^{N'} \beta_{is}^{(q)}, \tag{33}$$

where all draws come from the restricted sampler. These means are reported only when the corresponding posterior first moments are finite.

Joint hierarchical Gamma prior. The restricted hierarchical sampler updates each β_{is} from the truncated version of (24). The updates for b_s, a_s and α remain (25), (26) and (27), respectively, because the ordering indicator acts only on $\boldsymbol{\beta}$ and is constant in each of those conditional updates. The restriction is imposed on the β draws at every iteration. For this model, posterior means are calculated from the retained restricted posterior draws:

$$\widehat{\alpha}_{B,H,\mathcal{C}} = \frac{1}{N'} \sum_{q=1}^{N'} \alpha^{(q)}, \qquad \widehat{\beta}_{is,B,H,\mathcal{C}} = \frac{1}{N'} \sum_{q=1}^{N'} \beta_{is}^{(q)}. \tag{34}$$

The two order restrictions are treated separately; their simultaneous imposition is not considered in this paper.

4.4 HPD credible intervals

HPD intervals are calculated separately from the posterior samples generated under each prior specification. For either model, let $\eta^{(1)}, \ldots, \eta^{(N')}$ be the retained draws of a parameter or

reliability quantity, sorted as $\eta_{(1)} \le \cdots \le \eta_{(N')}$. For a $100(1-\gamma)\%$ interval, put $K = \lfloor (1-\gamma)N' \rfloor$. The sample HPD interval is the shortest interval

$$\left(\eta_{(l^*)}, \eta_{(l^*+K)}\right), \qquad l^* = \underset{1 \le l \le N'-K}{\arg\min} \left\{\eta_{(l+K)} - \eta_{(l)}\right\}. \tag{35}$$

Independent scale-invariant prior. In the unrestricted case, apply (35) to draws from the unrestricted sampler. In the order-restricted case, use only the retained draws from the restricted sampler described above. Calculate reliability quantities at each retained draw before applying the same shortest-interval procedure.

Joint hierarchical Gamma prior. Apply (35) separately to retained draws of α, each β_{is} and, when required, the hyperparameters a_s and b_s. Evaluate reliability quantities at each joint posterior draw before sorting. For order-restricted estimation, use only draws from the corresponding restricted posterior. Report the number of chains, warm-up and retained iterations, split-$\widehat{R}$, effective sample sizes and Monte Carlo standard errors alongside the HPD intervals.

5 Numerical Illustration

5.1 Simulation study

A simulation study will be conducted to compare the unrestricted and order-restricted MLEs with Bayesian estimates under the independent scale-invariant prior and the joint hierarchical Gamma model with informative hyperpriors. The study will include parameter settings for which the order restrictions are satisfied, violated, or partially satisfied, together with different sample sizes and progressive censoring plans. For each cause-specific lifetime, for each unit ℓ in group i, generate independent $U_{i\ell s} \sim \text{Uniform}(0,1)$ variables and transform them using

$$T_{i\ell s} = \left[1 - (1 - U_{i\ell s})^{1/\beta_{is}}\right]^{1/\alpha}, \qquad s = 1, 2.$$

The observed failure time for unit ℓ is $X_{i\ell} = \min(T_{i\ell 1}, T_{i\ell 2})$, and the observed cause is the cause attaining this minimum. The specified progressive Type-II removal plan is then applied to each group.

For a parameter θ, based on B simulated data sets, the absolute bias and mean-squared error will be calculated as

$$\text{AB}(\widehat{\theta}) = \left| \frac{1}{B} \sum_{b=1}^{B} \widehat{\theta}^{(b)} - \theta \right|, \qquad \text{MSE}(\widehat{\theta}) = \frac{1}{B} \sum_{b=1}^{B} (\widehat{\theta}^{(b)} - \theta)^2.$$

For intervals $[L_b, U_b]$, the empirical coverage probability and average width will be

$$\text{CP} = \frac{1}{B} \sum_{b=1}^{B} \mathbf{1}\{L_b \le \theta \le U_b\}, \qquad \text{AW} = \frac{1}{B} \sum_{b=1}^{B} (U_b - L_b).$$

The results will be arranged in tables to compare the classical estimators and both Bayesian prior specifications under unrestricted and order-restricted cases. For every Bayesian computation, the number of chains, warm-up and retained iterations, split-$\widehat{R}$, effective sample sizes and Monte

Carlo standard errors will be reported. The hierarchical prior hyperparameters will be stated explicitly, and sensitivity to their choices will be examined.

5.2 Real-data analysis

A data set will be selected only if its observations are compatible with the Kumaraswamy support $(0, 1)$ and the competing-risks structure. If a lifetime data set has a scientifically justified fixed upper mission time T_0, the transformed time $X = T/T_0$ may be used; T_0 must be specified from subject-matter considerations and not chosen as the observed sample maximum or by arbitrary min–max scaling. The source, groups, sample sizes, cause definitions, failure counts and censoring plan will be documented. If a complete data set is artificially subjected to a BPCS removal plan, it will be described as an induced-censoring illustration rather than as originally collected BPCS data. The fitted model and reliability estimates will be reported together with uncertainty intervals and suitable model-adequacy checks.

need to be done by harsh: complete the simulation study and real-data analysis, including the result tables and discussion of estimator performance.

6 Conclusion

A Kumaraswamy competing-risks model under block progressive censoring has been formulated with a common shape parameter and group- and cause-specific parameters. The likelihood leads to a one-dimensional profile problem for the unrestricted MLE, while the two separate order restrictions can be handled by PAVA or direct pooling. Bayesian estimation is developed under an independent scale-invariant prior and a joint hierarchical Gamma model with proper informative Gamma hyperpriors. Posterior means and HPD intervals are obtained from the corresponding posterior samples, both without and with order restrictions, provided posterior propriety and convergence conditions are verified. The simulation study and data analysis are required to assess the finite-sample performance and practical adequacy of the proposed methods.

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